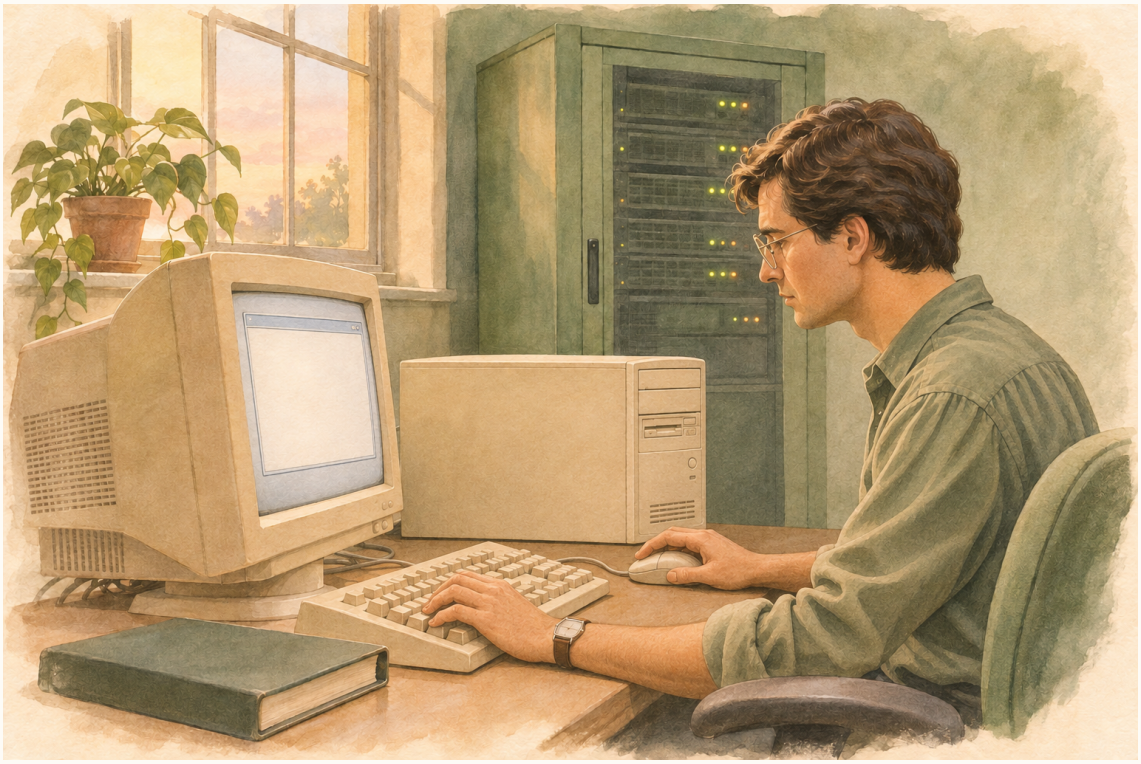
千年虫：被成功抹去的危机

Y2K: The Crisis That Success Erased



危机没有发生，能说明预防多余吗？

英文原文

The Y2K **bug**, or **millennium** bug, began as a **shortcut**. When **memory** and **hardware** were **costly**, **programmers** stored years as two **digits**, writing "99" for 1999. That **compact code** hid an **ambiguity**: a machine might read "00" as 1900, **corrupting** calculations of interest, ages, or **deadlines**.

Governments and firms **braced for chaos**. In February 1998 **President** Clinton created a **council** to **coordinate** America's effort. **Consultants audited legacy** systems and **patched faulty** date logic. In November 1999 the **Commerce Department** estimated American governments and **enterprises** would spend about $100bn; **projections** of the **global** bill ran to hundreds of billions.

Then **midnight struck** on 31 December 1999, and little went wrong. **Glitches** that did **crop up** were **sporadic** and mostly **minor**. Radiation-monitoring equipment **faltered** at Japan's Shika nuclear plant without any threat to safety, and a Pentagon ground system that processes **spy**-satellite data failed. In February 2000 a Senate committee **concluded** that the **expenditure** had been **justified**.

Many saw it differently: the threat had been **exaggerated**, even a **hoax**. Yet calm **prevailed largely** because so much had been fixed. This is the **preparedness paradox**: when **precautions** work, the danger they **ward off** never appears, so it looks **imaginary**. **Tedious** repairs earned **mockery** rather than **gratitude**.

Critics noted that Italy and **South Korea** spent far less yet **fared no worse**. The **comparison misleads**: those low figures often **omitted** what firms spent, and much of the **remedial** work replaced **obsolete** systems **overdue for** upgrading. Nobody can say which dollars were **essential** and which **excessive**. **Prevention** that works may leave nothing but an **invoice**, to **be weighed against** a **catastrophe** that might have been.

中文译文

Y2K 漏洞，又称千年虫，起初只是一条捷径。当年内存和硬件都很昂贵，程序员便把年份存成两位数字，用“99”表示1999年。这种小而紧凑的代码暗藏着一种歧义：机器可能把“00”读成1900年，从而搞错利息、年龄或截止日期的计算。

各国政府和企业都在严阵以待，准备迎接一场混乱。1998年2月，美国总统克林顿设立了一个委员会来协调全美的应对工作。顾问们审查了遗留系统，并给有缺陷的日期处理逻辑打上补丁。1999年11月，美国商务部估算，美国各级政府和企业将为此花费约1,000亿美元；对全球总账单的各种预测则高达数千亿美元。

随后，1999年12月31日的午夜钟声敲响，几乎没出什么问题。确实意外出现的小故障/小毛病也是零星的/偶发的，而且大多轻微。日本志贺核电站的辐射监测设备出了故障，但并未威胁安全；美国国防部一套处理间谍卫星数据的地面系统也曾失灵。2000年2月，美国参议院的一个委员会得出结论，认定这笔开支是合理的。

许多人却不这么看：在他们眼里，这一威胁被夸大了，甚至是一场骗局。然而，平静之所以占据上风，很大程度上正是因为大量问题早已修好。这就是防备悖论：预防措施一旦奏效，它们所抵御的危险就永远不会出现，于是看起来像是虚构的。那些枯燥乏味的修复工作换来的是嘲笑，而不是感激。

批评者指出，意大利和韩国花得少得多，结果也不比别国差。但这种比较具有误导性：那些低支出数字往往遗漏了企业自己的花费，而且很大一部分补救工作替换了早该升级的过时的/淘汰的系统。谁也说不清哪些钱是必不可少的，哪些是过度花费的。奏效的预防，可能只会留下一张账单，用来与一场本可能发生的灾难相权衡。

词汇与表达

bug n.
漏洞
The Y2K bug

millennium n.
千年

shortcut n.
捷径
began as a shortcut

memory n.
内存

hardware n.
硬件

costly adj.
昂贵

programmers n.
程序员

digits n.
数字

compact adj.
小而紧凑的

code n.
代码

ambiguity n.
歧义
That compact code hid an ambiguity

corrupting v.
搞错
corrupting calculations

deadlines n.
截止日期

braced for v.
严阵以待……准备迎接
firms braced for chaos

chaos n.
混乱

President n.
总统

council n.
委员会

coordinate v.
协调
to coordinate America's effort

Consultants n.
顾问

audited v.
审查了
Consultants audited legacy systems

legacy adj.
遗留
legacy systems

词汇与表达（续）

patched v.
给……打上补丁
patched faulty date logic

faulty adj.
有缺陷的

Commerce Department n.
美国商务部

enterprises n.
企业

projections n.
预测
projections of the global bill

global adj.
全球

midnight n.
午夜

struck v.
敲响

Glitches n.
小故障/小毛病

crop up phr.
意外出现

sporadic adj.
零星的/偶发的
sporadic and mostly minor

minor adj.
轻微

faltered v.
出了故障

spy n.
间谍

concluded v.
得出结论

expenditure n.
开支
the expenditure had been justified

justified adj.
合理的
the expenditure had been justified

exaggerated adj.
夸大了
the threat had been exaggerated

hoax n.
骗局
even a hoax

prevailed v.
占据上风

largely adv.
很大程度上

词汇与表达（续）

preparedness paradox n.
防备悖论
the preparedness paradox

precautions n.
预防措施
when precautions work

ward off phr.
抵御
the danger they ward off

imaginary adj.
虚构的

Tedious adj.
枯燥乏味的

mockery n.
嘲笑

gratitude n.
感激

South Korea n.
韩国

fared no worse phr.
不比别国差

comparison n.
比较

misleads v.
具有误导性

omitted v.
遗漏了
omitted what firms spent

remedial adj.
补救

obsolete adj.
过时的/淘汰的
replaced obsolete systems

overdue for adj.
早该

essential adj.
必不可少的
which dollars were essential

excessive adj.
过度花费的

Prevention n.
预防

invoice n.
账单

be weighed against phr.
与……相权衡

catastrophe n.
灾难
a catastrophe that might have been

文章简述

- 隐患：早期电脑为省内存，把1999年只记成“99”。跨入2000年时，“00”可能被当成1900年，导致年龄、利息和截止日期算错，这就是“千年虫”。
- 修补：各国政府和企业提前排查旧系统、修正日期处理。美国估算本国政府和企业总共要花约1000亿美元，全球预计数千亿美元。
- 跨年：2000年到来后，故障大多零星轻微。美国参议院委员会认为投入值得，也有人因此认为先前警告夸大，甚至是骗局。
- 反差：很多问题已提前修好，可能正是没出大事的重要原因；修得越成功，危险越不显眼，反而像原本不存在，这叫“防备悖论”。
- 权衡：不能只因灾难没发生就说钱白花了，要比较修复费用与不修可能造成的损失。但也不能据此断言每笔支出都必要；哪些钱确有用、哪些花多了，仍难逐笔确定。